

ChoreoFil: Dancing Filament Light Bulb

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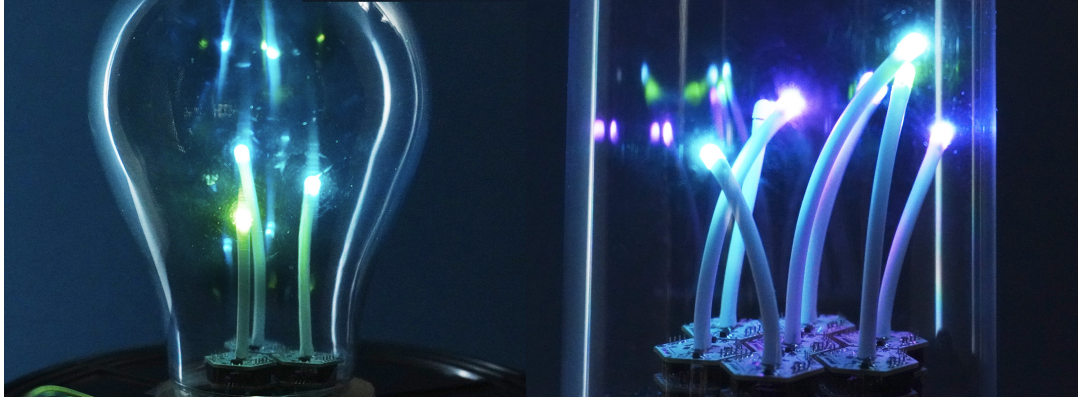


Figure 1: Snapshots of dancing filaments.

ABSTRACT

The ChoreoFil is a light bulb system with dancing filaments. The shape-memory alloy actuator as filaments bends like tentacles of a sea anemone, and the top of the actuator glows softly. By downsizing the previous system, the driving circuit could be fitted inside a light bulb. This project is still only an actuator-driven system, when connected to networks, information, and music, it will become the ambient media with an organism-like presence.

CCS CONCEPTS

• **Hardware** → **Sensors and actuators.**

KEYWORDS

Shape Changing Interface, Soft Actuator, Shape-Memory Alloy

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1 CHOREOFIL PROJECT

We have conducted several experiments and produced interactive art works [Nakayasu 2016, 2019] inspired by the waving tentacles of sea anemones. This research is a report on a light bulb system with

dancing filaments (Fig. 1). In previous projects completed as works of art, the movement of the actuators and the faint light create a comfortable impression on the viewer. The ChoreoFil project is a challenge to apply our technology to general-purpose products. The shape-memory alloy (SMA) actuator as a filament can bend in three directions and the top of the actuator glows softly. By downsizing the previous system, the circuit could be fitted inside a light bulb. The realization of implementation inside a light bulb shape makes it possible to easily replace existing light systems and expand the possibilities for spatial design. Moving filaments create an organic light effect and the movement of the light source produces not only light but also shadow wavering. The emotional light effects are expected to create a visually and psychologically immersive and comfortable feeling. Furthermore, the scalable system can be adapted to different numbers of actuators and light bulbs with different shapes (Fig. 2).

2 SYSTEM OVERVIEW

The ChoreoFil system is composed of an SMA actuator board, a microcontroller board, and a base board. The SMA actuator is composed of a silicon tube (3 mm in diameter), three BioMetal Fibers (a thin SMA wire developed by TOKI Corporation), a full colored LED chip, and a diffuser cap. The actuator can bend by the PWM control currents. And this system adopts the DMX 512, a serial communication protocol, which supports a daisy chain. The direction of bending, speed of movement, and light at the tip are controlled by DMX 512 signals. An SMA actuator board and a microcontroller board can be combined with different sized base board (Fig. 3). Therefore, it is possible to easily increase or decrease the number of actuators.

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Figure 2 : Light bulbs with different shapes

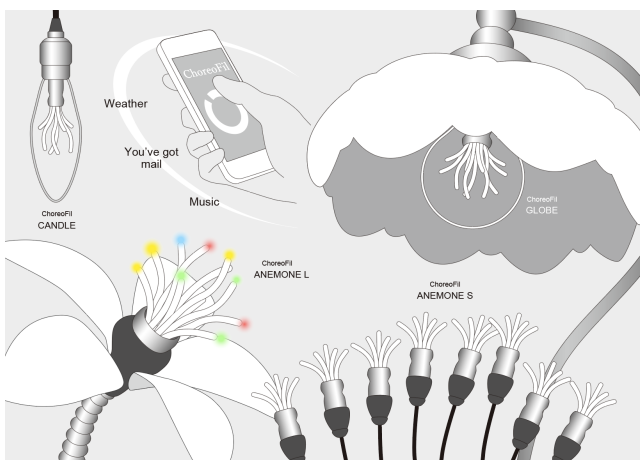


Figure 4 : A concept illustration of ambient media and various light bulb forms.

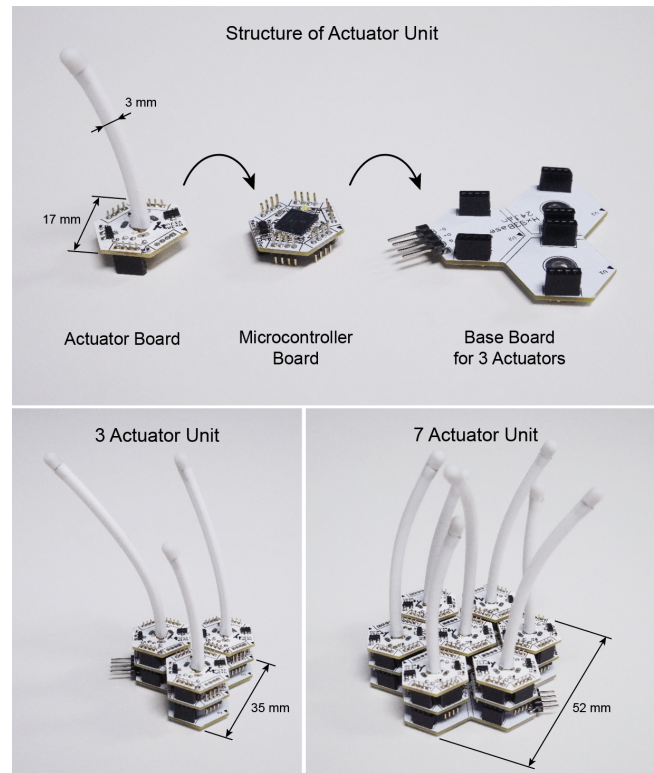


Figure 3 : Structure of actuator units

3 CONCEPT AS AMBIENT MEDIA

This project is still only an actuator-driven system, when connected to networks, information, and music, it will become the ambient media [Ishii 2008] with the presence of a living organism (Fig. 4). The ChoreoFil based ambient media allows users to receive information such as weather, incoming mail, etc. from a combination of light, movement and sound. It is not possible to communicate detailed information, but it allows subconscious access to rough information as if interacting with a living organism. Sometimes the filaments enjoy dancing with the music.

ACKNOWLEDGMENTS

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